

Does Class Size Matter?

A Century of Contested Evidence and What It Teaches Us About Causal Inference

Policy Context, Modeling Practice, and Meta-Analysis Translation

Key references: Glass & Smith (1979) · Shin & Chung (2009) · Hanushek & Krueger (2002)

1. The Basic Debate: A Policy Question With High Stakes

Class size reduction (CSR) is one of the most persistently contested questions in education policy — and one of the most expensive. Adding one teacher for every student removed from a classroom scales into billions of dollars at the national level. As Shin and Chung (2009) note, estimates of nationwide CSR program costs in the United States range from roughly \$2 billion to over \$11 billion per year in operational costs alone, before accounting for new construction, teacher pipeline constraints, and salary pressures.

The stakes are high in both directions. If class size genuinely matters, ignoring it condemns a generation of students — especially disadvantaged and minority students — to worse outcomes. If it doesn't, redirecting those resources toward teacher quality, curriculum, or targeted interventions may produce far greater gains per dollar spent. This is the core tension that runs through the research literature.

Despite over a century of study — Glass and Smith (1979) traced empirical work back to Rice (1902) — the literature accumulated in a state of near-paralysis. Review after review concluded that the evidence was unclear, contradictory, or simply inconclusive. Glass and Smith (1979) opened their landmark meta-analysis with a now-famous observation: the topic had "dissolved into cynical despair or epistemological confusion" and was treated as a dead issue by most instructional researchers.

Why Was the Literature So Confusing?

Previous narrative reviews were selective and haphazard in their literature searches, often ignoring dissertations and unpublished reports.

- Reviewers relied on informal 'vote counting' — tallying studies that found positive versus negative effects — without accounting for differences in effect magnitude, study quality, or class-size ranges compared.
- No standardized effect-size metric existed across studies, making quantitative aggregation impossible without methodological innovation.
- Studies ranged from comparing a class of 1 to a class of 40 (tutoring literature) to comparing classes of 28 versus 22 in a single district — wildly different interventions counted as equivalent evidence.

- Omitted variable bias and endogenous class assignment meant that naive comparisons were systematically biased in both directions depending on the institutional context.

2. Confounding Factors: Why the Relationship Is Difficult to Study

The central identification problem in class size research is that class size is not randomly assigned. Classrooms, schools, and districts that differ in class size differ in many other respects that independently affect student outcomes. These confounders push naive estimates in opposite directions, which is exactly why the raw literature appears so contradictory.

2.1 Positive Confounders (Upward Bias in Naive Estimates)

In the cross-section, class size is often positively associated with student test scores — meaning larger classes appear to produce better outcomes. This is counterintuitive but explainable:

- Higher-SES schools and districts tend to have larger enrollments and more students per classroom. Their students also score higher due to family background, parental education, and resources entirely unrelated to class size.
- More experienced teachers are often assigned to larger classes — either as a matter of district policy or because experienced teachers have seniority to choose higher-enrollment sections. Since experience is positively associated with teacher quality, this creates a spurious positive correlation between class size and achievement.
- Schools that add sections (and thus reduce class size) often do so in response to poor outcomes — struggling schools receive resources. This reverse causality means low class size and low performance become correlated in observational data.

2.2 Negative Confounders (Downward Bias or Over-Estimation of Effects)

Conversely, some study designs over-estimate the class size effect:

- Studies comparing individual tutoring (class size = 1) to typical classroom instruction (class size = 25–30) detect large effects, but these are driven largely by the qualitative shift in instructional modality, not class size per se.
- Short-term experiments may capture novelty or Hawthorne effects rather than sustained instructional benefits of smaller classes.
- Failure to control for pretest scores allows school-level SES to inflate the apparent class size effect when smaller classes are disproportionately assigned to lower-SES schools that receive policy interventions.

2.3 The Endogeneity of Class Assignment

Hanushek (2002) emphasizes that the history of the class size literature is largely a history of endogenous assignment. Pupil-teacher ratios fell by roughly one-third in the United States between 1960 and 1995 — a larger reduction than any policy currently proposed. Yet no corresponding improvement in aggregate student performance (measured by SAT scores or NAEP trends) is detectable over the same period. Hanushek argues this is the most powerful naturalistic test of the class size hypothesis: if the effect were large and general, *we should have seen it in the aggregate data*.

Krueger (2002) disputes this interpretation, noting that aggregate trends conflate many simultaneous changes — demographic shifts, changes in who takes the SAT, curriculum changes, and the composition of the teaching workforce. This methodological disagreement is itself instructive: the same historical data support opposite conclusions depending on the modeling assumptions brought to bear.

The Confounding Structure in the Synthetic Data

This synthetic dataset is constructed to reproduce the key confounding patterns found in real observational data:

- `free_lunch` (SES proxy) is negatively correlated with `mother_edu`, positively correlated with `class_size` in some schools, and negatively associated with pretest scores — mirroring the school-level SES gradient.
- `teacher_exp` is positively correlated with graduate degree attainment. Dropping either variable biases the other's coefficient.
- `class_size` is partly determined by `school_size_factor`, which is correlated with school SES — so OLS without fixed effects produces an endogenous estimate.
- pretest scores absorb much of the SES signal: models that include pretest will show weaker `free_lunch` and `mother_edu` coefficients than models that omit it.

3. How Meta-Analysis Untangled the Evidence

The methodological innovation that finally brought order to the class size literature was meta-analysis — the quantitative synthesis of effect sizes across studies. Glass and Smith (1979) introduced this approach to the class size literature in what became one of the most cited papers in education research. Their key contributions were:

1. A standardized effect size metric (Δ_{S-L}), defined as the mean achievement difference between a smaller and larger class divided by the within-class standard deviation. This allowed direct comparison across studies using different tests, subjects, and age groups.
2. A regression model of effect sizes as a function of the actual class sizes compared (S , S^2 , $L-S$), rather than crude categorical splits. This revealed the nonlinear shape of the class size–achievement relationship.
3. Systematic coding of study characteristics as potential moderators: year of study, grade level, subject, pupil IQ, duration of instruction, and crucially, the degree of experimental control (random assignment vs. matched vs. uncontrolled).

Working from 77 studies yielding 725 effect size comparisons — spanning nearly 900,000 pupils across 70 years of research in more than a dozen countries — Glass and Smith found a clear aggregate signal: on average, smaller classes outperformed larger ones, with the benefit concentrated at small class sizes. The effect was negligible moving from class size 30 to 40 but substantial moving from 20 to 10 or from 10 to 1.

3.1 What Moderates the Effect? Key Findings from the Meta-Analyses

Moderator	Glass & Smith (1979)	Shin & Chung (2009)
Overall effect size	$\Delta \approx 0.09$ SD average; up to 0.56 SD (class 1 vs. 40)	$d = 0.20$ SD (random-effects model)
Elementary vs. secondary	Secondary effect slightly stronger in bivariate; overlapping CIs	Elementary ($d=0.20$) clearly stronger than secondary ($d=-0.05$)
Grade-level trend	Not directly modeled	CSR effect decreases as grade increases; grades K–3 show strongest gains
Study quality	Well-controlled studies show much larger effects ($R=.621$) than poorly-controlled ($R=.187$)	Random-assignment studies show larger effects ($d=0.20$ vs. 0.11)
Publication status	Not directly examined	Published studies show larger effects ($d=0.21$) than unpublished ($d=0.13$)
Subject area	Effects do not differ appreciably across subjects	Reading ($d=0.19$) and math ($d=0.20$) positive; writing/SAT inconclusive
Pupil IQ / SES	No significant moderation by IQ	Larger effects for minority and disadvantaged students (cited from STAR)
Nonlinearity	Strong: marginal benefit greatest at small class sizes	Not directly modeled; linear assumption used

3.2 The Hanushek–Krueger Debate: A Lesson in Meta-Analytic Methodology

The Hanushek (2002) volume crystallizes a methodological debate that is directly relevant to how we interpret aggregated research. Hanushek's vote-counting approach counted each regression estimate from a published study as a separate vote, regardless of how many estimates came from the same study. Krueger challenged this on statistical grounds: studies that produce many estimates from the same data get disproportionate weight, and those multi-estimate studies tend to be of lower quality with more imprecise coefficients.

When Krueger re-weighted the literature to give each published study a single vote — the 'one study, one vote' principle — the aggregate evidence shifted substantially toward a positive class size effect. Hanushek disputed the weighting logic, arguing that each estimate is a distinct empirical exercise with independent information. This debate is unresolved but exposes a fundamental tension in research synthesis: the unit of analysis matters enormously.

Parallel in Our Modeling Exercise

This methodological dispute maps directly onto choices we will face with the synthetic panel data:

- If we pool all 1,248 classroom-year observations and estimate a single OLS model, each observation gets equal weight — analogous to Hanushek's approach.
- If we aggregate to the school level and run 8 school-level regressions, we give each 'study site' equal weight — analogous to Krueger's approach.
- These will produce different point estimates. Neither is definitively correct; each embeds assumptions about the independence structure of the data.
- Fixed effects (teacher_id, school_id, year) is a third approach that uses only within-unit variation — closest in spirit to a well-controlled experimental design.

4. The Post-Meta-Analysis Consensus (and Its Limits)

After Glass and Smith (1979) and the subsequent STAR experiment in Tennessee (1985–1990), a rough consensus emerged:

- Smaller classes do improve student achievement, but the effect is concentrated in the early grades (K–3) and is substantially larger for disadvantaged and minority students.
- The effect is nonlinear: reductions from 25 to 15 students produce more benefit per student moved than reductions from 35 to 25. Very small group sizes (under 10) show dramatically larger effects, but these blend into the tutoring literature.
- Effect sizes in well-controlled studies cluster around 0.15–0.25 SD for elementary grades — meaningful but not transformative. Krueger's cost-benefit analysis suggested the benefits of reducing K–3 classes from 22 to 15 students approximately break even at a 4% discount rate if the critical effect size of 0.10 SD is met.
- Teacher quality dominates class size as a determinant of student outcomes. Hanushek (2002) argues forcefully that the opportunity cost of CSR — hiring more teachers from a pool of marginal quality — may partially or fully offset the class size benefit.
- The effect is negligible or absent in secondary grades (9–12). Shin and Chung (2009) find $d = -0.05$ for secondary schools, though they note this estimate is based on very few effect sizes since most statewide CSR experiments focused on K–3.

An important qualification from Hanushek (2002): the evidence base is dominated by a single high-quality experiment (Project STAR in Tennessee), which accounted for 78 of 120 effect sizes in Shin and Chung's meta-analysis. This concentration of evidence creates a between-study dependence problem that inflates precision estimates. When Shin and Chung use state as the unit of analysis rather than individual effect size, the overall estimate drops from $d = 0.20$ to $d = 0.08$.

Policy Implications After the Meta-Analyses

- Class size reduction is cost-effective primarily in grades K–3, particularly for disadvantaged and minority students.
- Blanket CSR across all grades is almost certainly not cost-effective given the near-zero effects at secondary level.
- The opportunity cost argument (teacher quality) means CSR is most beneficial when a school or district can hire high-quality additional teachers — a constraint that often binds most tightly in the lowest-SES districts.
- California's experience (1996–97 CSR implementation) illustrates the risk: rapid class size reduction depleted experienced teachers from high-poverty urban schools as those teachers moved to newly opened suburban positions.

5. Translating Meta-Analytic Findings to Our Modeling Exercise

The synthetic dataset was designed to reproduce the major structural features of the class size literature in a controllable setting where the true data-generating process is known. This lets us demonstrate how different modeling choices recover — or distort — the true effects.

5.1 The Reversal Problem: Why Naive OLS Gets the Sign Wrong

In the raw data, regressing `test_score` on `class_size` without controls typically produces a positive coefficient. Larger classes appear to produce better outcomes. This is not an artifact of the simulation — it reproduces a pattern that frustrated narrative reviewers for decades.

The mechanism: schools with higher `free_lunch` rates (lower SES) tend to have smaller class sizes in this dataset, because lower-enrollment schools have smaller sections. Those same schools produce lower test scores due to SES factors that are entirely independent of class size. The SES gradient dominates the bivariate relationship.

Model	What It Tells Us
m1: <code>test_score ~ class_size</code>	Biased positive coefficient. Demonstrates the reversal problem.
m2: <code>+ free_lunch + mother_edu</code>	Adding SES controls flips the sign toward negative. Demonstrates omitted variable bias.
m3: <code>+ teacher_exp + student_aide + teacher_edu</code>	Teacher controls partially absorb confounding but miss the SES channel.
m5: full OLS with pretest + year FE	Absorbs most confounders. Coefficient approaches the true embedded parameter.
m7: feols with <code>teacher_id</code> + year FE	Within-teacher variation in class size. Cleanest observational estimate.

5.2 Grade-Level Heterogeneity: Fixed Effects vs. Interaction Models

One of the most robust findings from both meta-analyses is that class size matters more in earlier grades. Glass and Smith (1979) found stronger effects at secondary level in their

bivariate regressions, but this was confounded by the range of class sizes studied. Shin and Chung (2009), working with more recent experimental data, found the opposite: elementary effects ($d = 0.20$) were substantially larger than secondary effects ($d \approx 0$).

In the synthetic data, the class size effect is built in as a step function: -0.060 per student above 23 in grades K–2, -0.050 in grades 3–5, -0.040 in grades 6–8, and -0.030 in grades 9–12. This gradient is recoverable from the data — but the model specification matters enormously.

Two Ways to Model Grade-Level Heterogeneity

Model A — Grade Fixed Effects (parallel slopes):

$\text{test_score} \sim \text{grade_level} + \text{class_size} + \text{controls}$

Each grade gets its own intercept, but `class_size` has a SINGLE coefficient.

This constrains the class size effect to be identical in kindergarten and 12th grade.

The estimated coefficient is a weighted average of the true grade-specific effects.

It will appear smaller than the true K–2 effect and larger than the true 9–12 effect.

Model B — Grade $\times$ Class Size Interaction:

$\text{test_score} \sim \text{grade_level} * \text{class_size} + \text{controls}$

Each grade gets its own intercept AND its own `class_size` slope.

This allows the effect to vary — recovering the gradient built into the DGP.

The K–2 coefficient should be more negative; grades 9–12 should approach zero.

Interpretation: The 'overall' effect size from Model A is not wrong — it is a population-weighted average. But it obscures a finding with direct policy relevance: the effect is larger for part of the population (elementary) and negligible for another (secondary). Pooling without interaction leads to mis-specified resource allocation.

5.3 Comparing Effect Sizes Across Specifications

A central lesson of the meta-analysis literature is that reported effect sizes are a function of modeling choices, not just of the underlying phenomenon. The table below summarizes how the class size coefficient is expected to behave across the regression sequence in our modeling exercise.

Model	Expected <code>class_size</code> coefficient	Why
Naive OLS (no controls)	Positive ($\sim +0.10$ to $+0.30$)	SES confounding dominates; reversal problem
+ SES controls	Negative (~ -0.10 to -0.20)	SES channel blocked; true direction emerges

Model	Expected class_size coefficient	Why
+ Teacher controls only	Weakly negative or near zero	Teacher quality partially absorbs variation
+ Pretest + year FE	Negative (~ -0.15 to -0.25)	Most confounding removed; close to DGP parameters
Grade FE, pooled slope	Negative (~ -0.10 to -0.20)	Weighted average of grade-specific effects
Grade $\times$ Class Size interaction (K–2)	More negative (~ -0.25 to -0.35)	Recovers stronger true effect for early grades
Grade $\times$ Class Size interaction (9–12)	Near zero or weakly negative (~ 0 to -0.10)	Recovers weaker true effect for secondary
Teacher FE (feols)	Negative (~ -0.10 to -0.20)	Within-teacher variation; cleanest estimate

5.4 Interpreting Effect Size Magnitude

The synthetic test scores are centered at 50 with a standard deviation of approximately 12–13 points. An OLS coefficient on `class_size` of -0.15 means each additional student in a classroom costs about 0.15 test score points. Moving from a class of 15 to a class of 25 — a 10-student increase — costs approximately **1.5 points, or roughly 0.12 standard deviations**.

This is in the range of the Glass and Smith (1979) findings for class sizes in the 15–25 range ($\Delta \approx 0.05$ – 0.13 SD) and broadly consistent with the Shin and Chung (2009) overall meta-analytic estimate of $d = 0.20$ SD for elementary grades. The synthetic data is calibrated to be empirically plausible, not to reproduce any single study.

After adding the grade-level interaction, we should see that the K–2 effect is roughly twice as large as the pooled estimate, and the 9–12 effect is roughly half or less. This is the core policy finding of the post-meta-analysis consensus — and the reason why blanket CSR across all grades is difficult to justify on cost-effectiveness grounds.

6. Discussion Questions for the Modeling Exercise

1. The naive regression (`class_size` only) produces a positive coefficient. Before running the models, sketch out which variables you expect are driving this reversal. Which confounders would you control for first, and in what order?
2. Hanushek argued that pupil-teacher ratios fell steadily for 35 years in the U.S. with no detectable improvement in aggregate test scores. What omitted variables or confounders would need to be present to reconcile this historical pattern with a genuine positive effect of class size reduction?

3. When we add grade fixed effects to the model, the `class_size` coefficient changes. Is the fixed-effect model identifying a different quantity than the pooled model? What assumptions must hold for each?
4. Compare the coefficient on `class_size` from the grade-fixed-effects model to the average of the grade-specific interaction coefficients. Under what conditions would you expect these to differ? Under what weighting scheme would they be equal?
5. Shin and Chung (2009) found a publication bias: published studies showed larger effects ($d = 0.21$) than unpublished studies ($d = 0.13$). What mechanisms could produce this pattern? What does this imply for the 'true' population effect size?
6. The teacher fixed effects model (`feols`) identifies the class size effect from within-teacher variation across years. What is the identifying assumption? What threat to this assumption exists in our panel, and does it matter for interpretation?

7. Key References

Source	Key Contribution
Glass & Smith (1979) Ed. Evaluation & Policy Analysis 1(1)	First meta-analysis of class size literature. 77 studies, 725 effect sizes, 900,000 pupils. Introduced standardized effect size Δ and regression-based integration. Found strong nonlinear effect; negligible above class size 30, substantial below 20.
Shin & Chung (2009) KEDI Journal 6(2)	Updated meta-analysis of U.S. studies 1989–2008. 17 studies, 120 effect sizes. Random-effects overall $d = 0.20$ SD. Elementary schools ($d=0.20$) clearly outperform secondary ($d\approx 0$). Demonstrates dependence problem when STAR studies dominate.
Hanushek in Mishel & Rothstein, eds. (2002) The Class Size Debate, EPI	Argues vote-counting evidence is negative or mixed. Emphasizes that aggregate U.S. data (declining PTR, stagnant scores) contradicts a large general effect. Stresses teacher quality > class size. Historical review from 1960–1995.
Krueger in Mishel & Rothstein, eds. (2002) The Class Size Debate, EPI	Rebuts Hanushek's vote-counting methodology. One study, one vote re-weighting produces positive aggregate result. Cost-benefit analysis suggests K–3 CSR from 22 to 15 students approximately breaks even at 4% discount rate if effect ≥ 0.10 SD.
Tennessee STAR Experiment (1985–1990)	Gold-standard RCT. Students/teachers randomly assigned to small (13–17), regular (22–26), or regular+aide classes in grades K–3. Small class advantage ≈ 0.20 – 0.25 SD; minority students showed effects roughly double those of white students. Most cited single study in the literature.

Simulation Note: The synthetic panel dataset was constructed to embed the key structural relationships identified in this literature. The data-generating process is fully observable in the R code, allowing students to verify their empirical estimates against known parameters — a pedagogical advantage unavailable with real data.